TSSM
Creating VCE Success

2025 Trial Examination

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PROCESSING LABEL HERE

Write your **student number** in the boxes above. **Letter**

Specialist Mathematics
Written Examination 1

Units 3 & 4 – Question and Answer Book

VCE Examination

Reading time: **15 minutes**
Writing time: **1 hour**

Materials supplied

- Task book of 13 pages

Instructions

- Write your responses in English.
- Answer **all** questions in booklet provided.
- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers.
- Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices, notes of any kind, blank sheets of paper and/or white out liquid/tape into the examination room.
- No calculator is permitted in this examination.
- Unless otherwise specified, an **exact** answer is required to a question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this booklet are not drawn to scale.
- Take the acceleration due to gravity to have magnitude $g\ \mathrm{ms}^{-2}$, where $g = 9.8$.

Contents pages

(9 questions, 40 marks)_____ 2-13

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Question 1 (3 marks)

The percentage (P) of a particular radioactive substance remaining after t years is reducing according to the differential equation:

$\frac{dP}{dt} = kP$ where t is in years.

Show that the solution to the differential equation is of the form: $P = 100e^{kt}$

where $k = -nlog_e 2$

Hence find the value of n given that 25% of the substance remains radioactive after 200 years.

3 marks

Question 2 (7 marks)

Consider the equation: $z^4 = x + yi,\ x, y \in R,\ z \in C$ where $z_1 = 1 + \sqrt{3}\, i$ is one of the roots of the equation.

a. Write z_1 in polar form.

1 mark

b. Hence, or otherwise, write in polar form, z_2, z_3, z_4 the other three roots of
$z^4 = x + yi, \qquad z \in C$

1 mark

c. Find the values of x and y.

2 marks

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d. Write $\sqrt[8]{x+yi}$ in simplified rectangular form.

3 marks

1 + 1 + 2 + 3 = 7 marks

Question 3 (3 marks)

Consider the number n where $n = \log_2 3$.

By assuming that n is a rational number, prove by contradiction that n is irrational.

3 marks

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Question 4 (4 marks)

Find the real values, a and b such that: $\int_0^{\sqrt{3}} (\tan^{-1} x)\, dx = a\pi - \log_e b$.

4 marks

Question 5 (3 marks)

Find the gradient of the tangent to the curve defined by the equation: $4x^3y + y^2 - x - 11 = 0$ at the point $(1,2)$.

3 marks

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Question 6 (5 marks)

The following approximations may be useful for this question.

Let Z be an observation from a standard normal distribution.

$\Pr(-1 \le Z \le 1) \approx 0.683, \Pr(-2 \le Z \le 2) \approx 0.954,\ \Pr(-3 \le Z \le 3) \approx 0.997$

POTTY POTS produce a large handmade ceramic garden pot. (Called an OASIS.) Variation in weight between pots of the 'same size' occurs.

The weight of these pots is normally distributed with a mean of $6.6kg$ and a variance of $0.45kg$.

Brigette orders 5 OASIS pots which are dispatched in a wooden container weighing $3.2\ kg$.

Let the random variable W be the total weight of these five pots plus the wooden container.

a. Find $E(W)$ and $sd(W)$.

3 marks

b. Find, correct to two decimal places, the chance that Brigette's five pots (plus wooden container) are heavier than $33.2kg$ but lighter than $37.7kg$.

2 marks

3 + 2 = 5 marks

Question 7 (6 marks)

Consider the function: $f: (-1,5] \to R \; where \; f(x) = \frac{1}{2}x^3(x-4)$

a. State the coordinates of the stationary point of inflection for $y = f(x)$.

_____

_____

1 mark

b. Find the coordinates of the non-stationary point of inflection for $y = f(x)$ hence find the maximum negative gradient of this function. (Call this point N.)

_____

_____

_____

_____

2 marks

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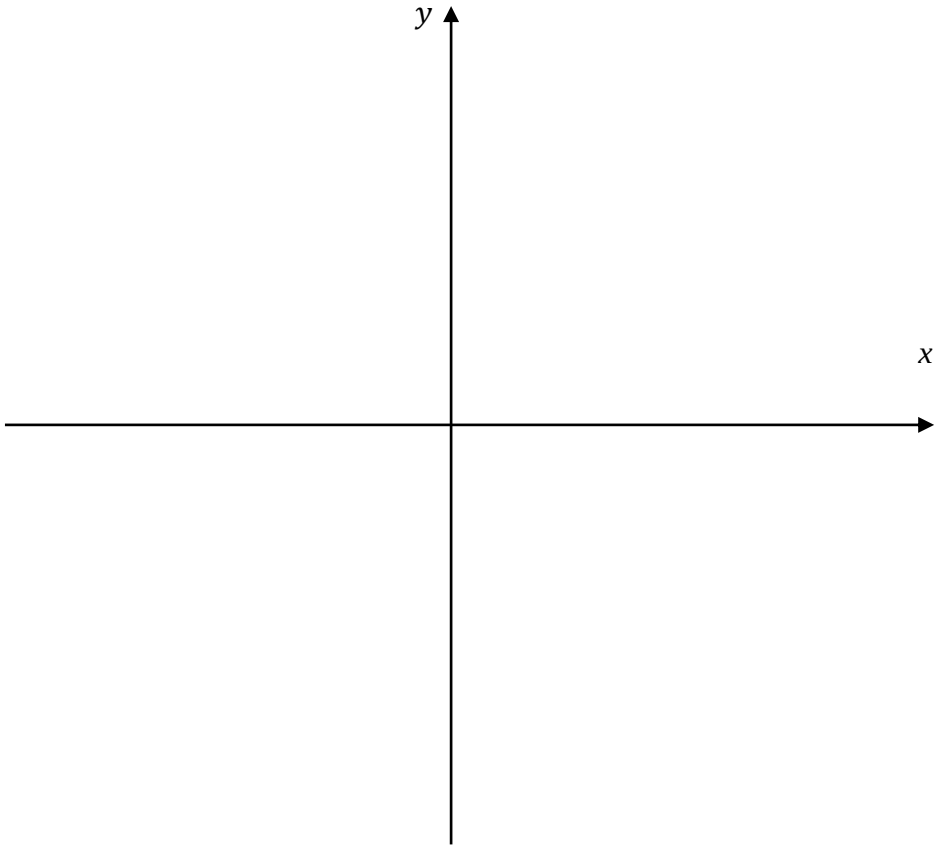
c. Sketch $y = f(x)$ and $y = \frac{1}{f(x)}$ on the same axes. Label any asymptotes, point N, stationary points and end points. (Your sketch will provide an approximate scale only in the vertical direction.)



3 marks

1 + 2 + 3 = 6 marks

Question 8 (3 marks)

Consider the function: $f: [0, \frac{\pi}{4}] \to R$ where $f(x) = \cos 2x$

Find the volume formed when the area bound by $f(x) = \cos 2x$, the x axis and the y axis is rotated around the x axis.

3 marks

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Question 9 (6 marks)

Consider the points in three dimensional space: $P(1,1,2), Q(-1,2,7), R(0,-1,2)$.

a. Find $\overrightarrow{PQ} \times \overrightarrow{PR}$, the cross product of $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

2 marks

b. Hence find a unit vector normal to the plane containing points P, Q, R.

1 mark

c. Find the distance of point $S = (3, -1, 2)$ from the plane containing the points P, Q, R.

3 marks

2 + 1 + 3 = 6 marks

End of examination questions

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